

考研竞赛凯哥 第7次周考

考试时间: 4月26日上午08:30~10:00

讲解时间: 4月27日上午08:30~10:00

注1: 本次周考的范围为《极限与连续》《一元微分的计算、证明》《一元积分的计算、证明、应用》

注2: 周考, 每周一次, 比较频繁, 所以题量减半, 考试时间减半, 且选填题与解答题的比例随机;

月考, 每月一次, 完全按照考研的标准进行设置.

注3: 无论周考还是月考, 目的都是查漏补缺和巩固所学知识, 所以题目的难度至少持平或略高于真题的平均难度, 大家一定不要因为模考成绩不理想而灰心, 考前犯的任何错误都应该珍惜!

1. $\lim_{n \rightarrow \infty} \frac{1}{n} \left(\frac{1}{n+1} + \frac{2}{n+2} + \frac{3}{n+3} + \cdots + \frac{3n}{4n} \right).$

解: $I = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^{3n} \frac{k}{n+k} = \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{k=1}^{3n} \frac{\frac{k}{n}}{1 + \frac{k}{n}} = \int_0^3 \frac{x}{1+x} dx = \int_0^3 \left(1 - \frac{1}{1+x} \right) dx = 3 - \ln 4.$

2. $\int \frac{1}{1 + \sin x + \cos x} dx$

解: 用万能公式, 令 $\tan \frac{x}{2} = t$, 得 $\sin x = \frac{2t}{1+t^2}$, $\cos x = \frac{1-t^2}{1+t^2}$, $x = 2 \arctan t$, $dx = \frac{2}{1+t^2} dt$, 代入得

$$I = \int \frac{1}{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}} \cdot \frac{2}{1+t^2} dt = 2 \int \frac{1}{1+t^2+2t+1-t^2} dt = \int \frac{1}{1+t} dt = \ln|1+t| + C = \ln \left| 1 + \tan \frac{x}{2} \right| + C$$

3. $\int_0^{+\infty} \frac{1}{(1+x^2)(1+x^a)} dx$

解: 倒代换后, 和原积分相加即可. $I \stackrel{x=\frac{1}{t}}{=} \int_{+\infty}^0 \frac{1}{\left(1+\frac{1}{t^2}\right)\left(1+\frac{1}{t^a}\right)} \left(-\frac{1}{t^2} dt\right) = \int_0^{+\infty} \frac{t^a}{(t^2+1)(t^a+1)} dt.$

故 $I = \frac{1}{2} \int_0^{+\infty} \frac{1+x^a}{(1+x^2)(1+x^a)} dx = \frac{1}{2} \int_0^{+\infty} \frac{1}{1+x^2} dx = \frac{1}{2} \arctan x \Big|_0^{+\infty} = \frac{\pi}{4}.$

4. (1) $\int_0^{\frac{\pi}{2}} \frac{\sin x \cos x}{\sin x + \cos x} dx;$

(2) $\int_0^{\frac{\pi}{2}} \frac{\sin x \cos^2 x}{\sin x + \cos x} dx$

(3) $\int_0^{\frac{\pi}{2}} \frac{\sin x \cos^3 x}{\sin x + \cos x} dx$

解: (1) $I_1 = \int_0^{\frac{\pi}{2}} \frac{\frac{(\sin x + \cos x)^2 - 1}{2}}{\sin x + \cos x} dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} \left(\sin x + \cos x - \frac{1}{\sin x + \cos x} \right) dx = 1 - \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{1}{\sin x + \cos x} dx$

$$= 1 - \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{1}{\sqrt{2} \sin \left(x + \frac{\pi}{4} \right)} dx = 1 - \frac{1}{2\sqrt{2}} \ln \left| \csc \left(x + \frac{\pi}{4} \right) - \cot \left(x + \frac{\pi}{4} \right) \right| \Big|_0^{\frac{\pi}{2}} = 1 - \frac{1}{2\sqrt{2}} \ln \left| \frac{\sqrt{2} - (-1)}{\sqrt{2} - 1} \right|$$

$$= 1 - \frac{1}{2\sqrt{2}} \ln \frac{\sqrt{2}+1}{\sqrt{2}-1} = 1 - \frac{\sqrt{2}}{2} \ln(\sqrt{2}+1);$$

$$(2) \text{ 区间再现. } I_2 = \int_0^{\frac{\pi}{2}} \frac{\sin x \cos^2 x}{\sin x + \cos x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos x \sin^2 x}{\sin x + \cos x} dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{\sin x \cos x (\cos x + \sin x)}{\sin x + \cos x} dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \sin x \cos x dx = \frac{1}{4} \sin^2 x \Big|_0^{\frac{\pi}{2}} = \frac{1}{4}.$$

$$(3) \text{ 区间再现. } I_3 = \int_0^{\frac{\pi}{2}} \frac{\sin x \cos^3 x}{\sin x + \cos x} dx = \int_0^{\frac{\pi}{2}} \frac{\cos x \sin^3 x}{\sin x + \cos x} dx = \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{\sin x \cos x (\cos^2 x + \sin^2 x)}{\sin x + \cos x} dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} \frac{\sin x \cos x}{\sin x + \cos x} dx = \frac{1}{2} I_1 = \frac{1}{2} - \frac{\sqrt{2}}{4} \ln(\sqrt{2}+1).$$

5. 设 $g(x) = \begin{cases} x^3 \sin \frac{1}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases}$, $f(x)$ 可导, $F(x) = f[g(x)]$, 求 $F'(0)$.

解: $F'(0) = f'[g(0)] \cdot g'(0) = f'(0)g'(0)$.

其中 $g'(0) = \lim_{x \rightarrow 0} \frac{g(x) - g(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{x^3 \sin \frac{1}{x} - 0}{x - 0} = \lim_{x \rightarrow 0} x^2 \sin \frac{1}{x} = 0$, 故 $F'(0) = f'(0)g'(0) = 0$.

注: 本题很特殊, 不能用导数定义! 我们来看看如果用导数定义, 哪里会出错——

$$F'(0) = \lim_{x \rightarrow 0} \frac{F(x) - F(0)}{x - 0} = \lim_{x \rightarrow 0} \frac{f\left(x^3 \sin \frac{1}{x}\right) - f(0)}{x - 0} \stackrel{(*)}{=} \lim_{x \rightarrow 0} \frac{f\left(x^3 \sin \frac{1}{x}\right) - f(0)}{x^3 \sin \frac{1}{x}} \cdot \frac{x^3 \sin \frac{1}{x}}{x} = f'(0) \cdot 0 = 0$$

问题就出在上面的(*)处, 分母 $x^3 \sin \frac{1}{x}$ 在 $x=0$ 的去心邻域内有无穷多个零点!

这导致 $x \rightarrow 0$ 时, 函数 $\frac{f\left(x^3 \sin \frac{1}{x}\right) - f(0)}{x^3 \sin \frac{1}{x}}$ 有无穷多个无定义点, 在考研大纲的范围内, 无法求极限!

这和凯哥讲的“ $\lim_{x \rightarrow 0} \frac{\sin\left(x^2 \sin \frac{1}{x}\right)}{x^2 \sin \frac{1}{x}}$ 属于不能求极限, 而不是常规意义上的极限不存在”是一样的道理.

6. 曲线 $y = e^x$ 与 $x + y = e + 1$ 及 y 轴所围成的图形为 D , 求 D 绕直线 $y = e$ 旋转一周所得的旋转体体积.

解: (自己画图)

本题套公式即可, 但需要注意, 下边界 $y = e^x$ 绕着 $y = e$ 旋转得到旋转体, 把上边界 $x + y = e + 1$ 绕着 $y = e$ 旋转得到的旋转体完全包裹 (覆盖) 住了, 所以不需要重复计算!

故——
$$V = \int_0^1 \pi r^2 dx = \int_0^1 \pi (e - e^x)^2 dx = \frac{\pi}{2} (4e - e^2 - 1).$$

7. 设 $f(x) = x + \ln(2-x)$.

(1) 求 $f(x)$ 的最大值;

(2) 设 $a_1 = \ln 2$, $a_{n+1} = f(a_n)$, 证明 $\lim_{n \rightarrow \infty} a_n$ 存在, 并求该极限.

解: (1) 定义域为 $x < 2$. $f'(x) = 1 - \frac{1}{2-x} = \frac{1-x}{2-x}$, 故 $f(x)$ 在 $(-\infty, 1)$ 单调递增, 在 $(1, 2)$ 单调递减, 故 $f(x)$ 的最大值为 $f(1) = 1$.

(2) 有界性

1) $a_1 = \ln 2 < 1$;

2) 假设 $a_n < 1$;

3) 由于 $f(x)$ 的唯一最小值点为 $x=1$, 且 $f(1)=1$, 故当 $a_n < 1$ 时, $a_{n+1} = f(a_n) < 1$.

由数学归纳法可得, $a_n < 1$ 对一切的正整数 n 均成立, 即 $\{a_n\}$ 有上界.

单调性

$a_{n+1} - a_n = f(a_n) - a_n = \ln(2 - a_n)$, 由于 $a_n < 1$, 故 $\ln(2 - a_n) > 0$, $a_{n+1} > a_n$, 即 $\{a_n\}$ 单调递增.

由单调有解准则, $\{a_n\}$ 收敛.

设 $\lim_{n \rightarrow \infty} a_n = A$, 则 $A = f(A)$, 即 $A = A + \ln(2 - A)$, 解得唯一解 $A = 1$, 故 $\lim_{n \rightarrow \infty} a_n = 1$.

8. $\lim_{x \rightarrow +\infty} \frac{\int_0^x t \sin^2 t dt}{x^2}.$

解: 本题和上周的周考题 “ $\lim_{x \rightarrow +\infty} \frac{\int_0^x |\cos t| dt}{x}$ ” 非常相似, 解法一模一样!

设 $S(x) = \int_0^x t \sin^2 t dt$, $S'(x) = x \sin^2 x \geq 0$ 在 $x > 0$ 时恒成立, 且等号仅在 $x = n\pi$ 取到,

故 $S(x)$ 在 $(0, +\infty)$ 严格单调递增, 故当 $n\pi \leq x < (n+1)\pi$ 时, 有 $S(n\pi) \leq S(x) < S[(n+1)\pi]$.

$$\begin{aligned} S(n\pi) &= \int_0^{n\pi} x \sin^2 x dx = \int_0^{n\pi} (n\pi - x) \sin^2 (n\pi - x) dx = \int_0^{n\pi} (n\pi - x) \sin^2 x dx = \frac{n\pi}{2} \int_0^{n\pi} \sin^2 x dx \\ &= \frac{n\pi}{2} \cdot n \int_0^\pi \sin^2 x dx = \frac{n\pi}{2} \cdot n \cdot 2 \int_0^{\frac{\pi}{2}} \sin^2 x dx = n^2 \pi \frac{1}{2} \frac{\pi}{2} = \frac{n^2 \pi^2}{4}. \end{aligned}$$

同理, $S[(n+1)\pi] = \frac{(n+1)^2 \pi^2}{4}$, 故 $\frac{n^2 \pi^2}{4} \leq S(x) < \frac{(n+1)^2 \pi^2}{4}$.

$$\text{由 } \begin{cases} n\pi \leq x < (n+1)\pi \\ \frac{n^2 \pi^2}{4} \leq S(x) < \frac{(n+1)^2 \pi^2}{4} \end{cases} \text{ 可得, } \frac{\frac{n^2 \pi^2}{4}}{(n+1)^2 \pi^2} \leq \frac{S(x)}{x^2} \leq \frac{\frac{(n+1)^2 \pi^2}{4}}{n^2 \pi^2}.$$

$\lim_{n \rightarrow \infty} \frac{\frac{n^2 \pi^2}{4}}{(n+1)^2 \pi^2} = \frac{1}{4}$, $\lim_{n \rightarrow \infty} \frac{\frac{(n+1)^2 \pi^2}{4}}{n^2 \pi^2} = \frac{1}{4}$ 又由于本题中 $x \rightarrow +\infty$ 与 $n \rightarrow \infty$ 为充要条件,

故在 $\frac{\frac{n^2\pi^2}{4}}{(n+1)^2\pi^2} \leq \frac{S(x)}{x^2} \leq \frac{\frac{(n+1)^2\pi^2}{4}}{n^2\pi^2}$ 中令 $x \rightarrow +\infty$ (此时也必有 $n \rightarrow \infty$),

由夹逼准则可得, $\lim_{x \rightarrow +\infty} \frac{S(x)}{x^2} = \frac{1}{4}$, 即 $\lim_{x \rightarrow +\infty} \frac{\int_0^x t \sin^2 t dt}{x^2} = \frac{1}{4}$.

9. 设 $f(x)$ 在 $[0, 1]$ 上二阶连续可导, $f\left(\frac{1}{2}\right) = 1$, $\int_0^1 f(x) dx = \frac{5}{4}$, 证明: $\exists \xi \in (0, 1)$, 使得 $f''(\xi) = 6$.

解: 令 $F(x) = \int_0^x f(t) dt$, 则 $F(x)$ 三阶连续可导, 且 $F(0) = 0$, $F'\left(\frac{1}{2}\right) = 1$, $F(1) = \frac{5}{4}$.

将 $F(0)$ 和 $F(1)$ 在 $x = \frac{1}{2}$ 处展开为带拉格朗日余项的二阶泰勒公式, 得——

$$F(0) = F\left(\frac{1}{2}\right) + F'\left(\frac{1}{2}\right)\left(0 - \frac{1}{2}\right) + \frac{F''\left(\frac{1}{2}\right)}{2!}\left(0 - \frac{1}{2}\right)^2 + \frac{F'''(\xi_1)}{3!}\left(0 - \frac{1}{2}\right)^3 \quad ①$$

$$F(1) = F\left(\frac{1}{2}\right) + F'\left(\frac{1}{2}\right)\left(1 - \frac{1}{2}\right) + \frac{F''\left(\frac{1}{2}\right)}{2!}\left(1 - \frac{1}{2}\right)^2 + \frac{F'''(\xi_2)}{3!}\left(1 - \frac{1}{2}\right)^3 \quad ②$$

② - ① 得, $F(1) - F(0) = F'\left(\frac{1}{2}\right) + \frac{F'''(\xi_2) + F'''(\xi_1)}{48}$, 将 $F(0) = 0$, $F'\left(\frac{1}{2}\right) = 1$, $F(1) = \frac{5}{4}$ 代入得,

$$\frac{5}{4} = 1 + \frac{F'''(\xi_2) + F'''(\xi_1)}{48}, \text{ 故 } 6 = \frac{F'''(\xi_2) + F'''(\xi_1)}{2}.$$

由于 $F'''(x)$ 连续, 由介值定理可得, 存在 $\xi \in (\xi_1, \xi_2) \subset (0, 1)$, 使得 $F'''(\xi) = \frac{F'''(\xi_2) + F'''(\xi_1)}{2}$.

故 $F'''(\xi) = 6$, 即 $f''(\xi) = 6$, 证毕.

10. 设 $f(x) = \frac{1}{\pi x} + \frac{1}{(1-x)\sin\pi x} - \frac{1}{\pi(1-x)^2}$, $x \in \left[\frac{1}{2}, 1\right)$.

(1) 如何补充定义 $f(1)$, 才能使得 $f(x)$ 在 $\left[\frac{1}{2}, 1\right]$ 上连续;

(2) 在(1)的补充定义下, 求 $f'(1)$.

解: (1) $\lim_{x \rightarrow 1^-} f(x) = \lim_{x \rightarrow 1^-} \left[\frac{1}{\pi x} + \frac{1}{(1-x)\sin\pi x} - \frac{1}{\pi(1-x)^2} \right] = \frac{1}{\pi} + \lim_{x \rightarrow 1^-} \frac{\pi(1-x) - \sin\pi x}{\pi(1-x)^2 \sin\pi x}$
 $= \frac{1}{\pi} + \lim_{x \rightarrow 1^-} \frac{\pi(1-x) - \sin\pi x}{\pi(1-x)^2 \sin(\pi - \pi x)} = \frac{1}{\pi} + \lim_{x \rightarrow 1^-} \frac{\pi(1-x) - \sin[\pi(1-x)]}{\pi^2(1-x)^3} \xrightarrow{1-x=t} \frac{1}{\pi} + \lim_{x \rightarrow 0^+} \frac{\pi t - \sin(\pi t)}{\pi^2 t^3}$
 $= \frac{1}{\pi} + \lim_{x \rightarrow 0^+} \frac{\frac{1}{6}(\pi t)^3}{\pi^2 t^3} = \frac{1}{\pi} + \frac{\pi}{6}.$

故补充定义, 令 $f(1) = \frac{1}{\pi} + \frac{\pi}{6}$, 才能使得 $f(x)$ 在 $\left[\frac{1}{2}, 1\right]$ 上连续.

$$\begin{aligned}
 (2) \quad f'_-(1) &= \lim_{x \rightarrow 1^-} \frac{f(x) - f(1)}{x - 1} = \lim_{x \rightarrow 1^-} \frac{\frac{1}{\pi x} + \frac{1}{(1-x)\sin \pi x} - \frac{1}{\pi(1-x)^2} - \left(\frac{1}{\pi} + \frac{\pi}{6}\right)}{x - 1} \\
 &= \lim_{x \rightarrow 1^-} \frac{\left(\frac{1}{\pi x} - \frac{1}{\pi}\right)}{x - 1} + \lim_{x \rightarrow 1^-} \frac{\frac{1}{(1-x)\sin \pi x} - \frac{1}{\pi(1-x)^2} - \frac{\pi}{6}}{x - 1} \\
 &= \frac{1}{\pi} \lim_{x \rightarrow 1^-} \frac{\left(\frac{1}{x} - 1\right)}{x - 1} + \lim_{x \rightarrow 1^-} \frac{\frac{1}{(1-x)\sin(\pi - \pi x)} - \frac{1}{\pi(1-x)^2} - \frac{\pi}{6}}{x - 1} \\
 &= -\frac{1}{\pi} - \lim_{t \rightarrow 0^+} \frac{\frac{1}{t \sin(\pi t)} - \frac{1}{\pi t^2} - \frac{\pi}{6}}{t} \\
 &= -\frac{1}{\pi} - \lim_{t \rightarrow 0^+} \frac{\frac{\pi t - \sin(\pi t) - \frac{\pi}{6} \pi t^2 \sin(\pi t)}{\pi t^2 \sin(\pi t)}}{t} \\
 &= -\frac{1}{\pi} - \lim_{t \rightarrow 0^+} \frac{\pi t - \sin(\pi t) - \frac{\pi}{6} \cdot \pi t^2 \sin(\pi t)}{\pi^2 t^4} \\
 &= -\frac{1}{\pi} - \lim_{t \rightarrow 0^+} \frac{\pi t - \left[(\pi t) - \frac{1}{6}(\pi t)^3 + o(t^4)\right] - \frac{\pi}{6} \cdot \pi t^2 \left[(\pi t) - \frac{1}{6}(\pi t)^3 + o(t^4)\right]}{\pi^2 t^4} \\
 &= -\frac{1}{\pi} - \lim_{t \rightarrow 0^+} \frac{o(t^4)}{\pi^2 t^4} = -\frac{1}{\pi}.
 \end{aligned}$$